

Introduction

- Explosion of massive star (explosion of progenitor with $M > 8 M_{\odot}$).
- The core of these stars collapse under their own gravity (emission of thermal neutrinos (MeV)) over a time scale of few seconds.
- Some CCSNe are good candidates for the production of high-energy (HE) neutrinos.
<https://arxiv.org/abs/2206.00450>

GeV-TeV neutrinos through the interaction of ejecta with circumstellar material (CSM).
TeV-PeV neutrinos through choked jets (CJs).

Motivations

- Increase the possibility to observe a CCSN.
- Give insight into the cosmic-ray acceleration processes.

HE CCSN neutrino production mechanisms

HE CCSN neutrino production (theoretical models):

- **Relativistic Chocked jets (GeV-TeV)** jets of material trapped in the outer shells of the star. Only high energy neutrinos are able to escape these jets
Neutrino emission times between 10 s to 10⁴ s.

- **Interaction of the supernova ejecta and a dense circumstellar material (CSM) (TeV-PeV)**
Neutrino emission times between 2 hours 1 year post-core bounce.

$$pp \rightarrow \pi^+ \rightarrow \mu^+ \nu_\mu \rightarrow e^+ \nu_e \nu_\mu \bar{\nu}_\mu$$

CSM is characterised by its' **density profile**:

$$\rho_{cs} = Dr^{-w}$$

w wind profile ($w = 2$ reasonable in many cases)

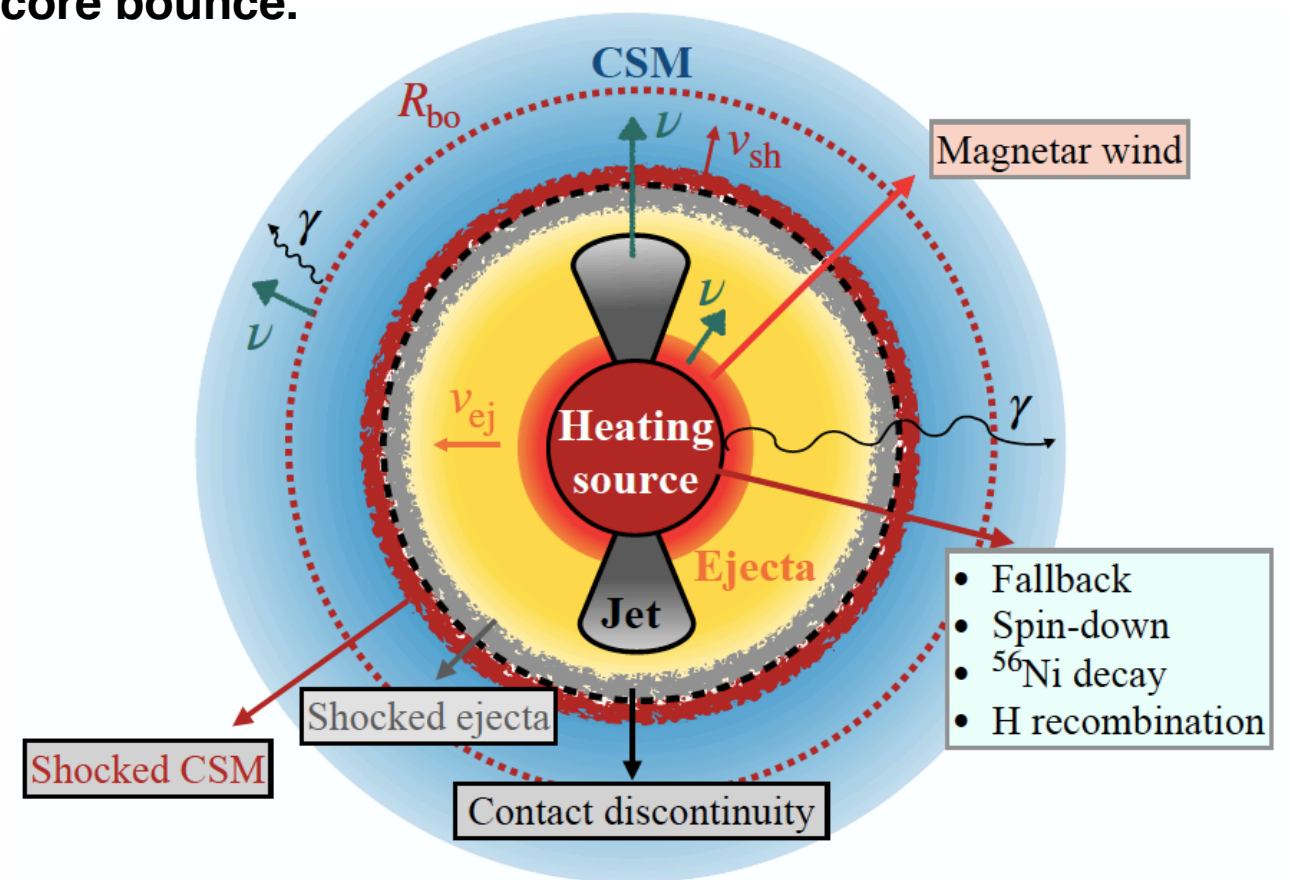
In the wind case: $D = 5 \times 10^{16} \text{ g cm}^{-1} D_*$

$$D = \frac{\dot{M}_w}{4\pi V_w}, \text{ with } \dot{M}_w \text{ the mass loss rates and } V_w \text{ the wind velocity}$$

CSM parameters for various types of SNe

Class	D_*	$\dot{M}_w [M_\odot \text{ yr}^{-1}]$	$V_w [\text{km s}^{-1}]$	$R_* [\text{cm}]$
IIn	1	10^{-1}	100	10^{13}
II-P ^a	10^{-2}	10^{-3}	100	6×10^{13}
II-P ^b	1.34×10^{-4}	2×10^{-6}	15	6×10^{13}

[Murase, arXiv:1705.04750 \[astro-ph.HE\]](#)



Key parameters affecting the HE ν flux

- mass loss
- wind velocity
- shock velocity
- proton spectral index

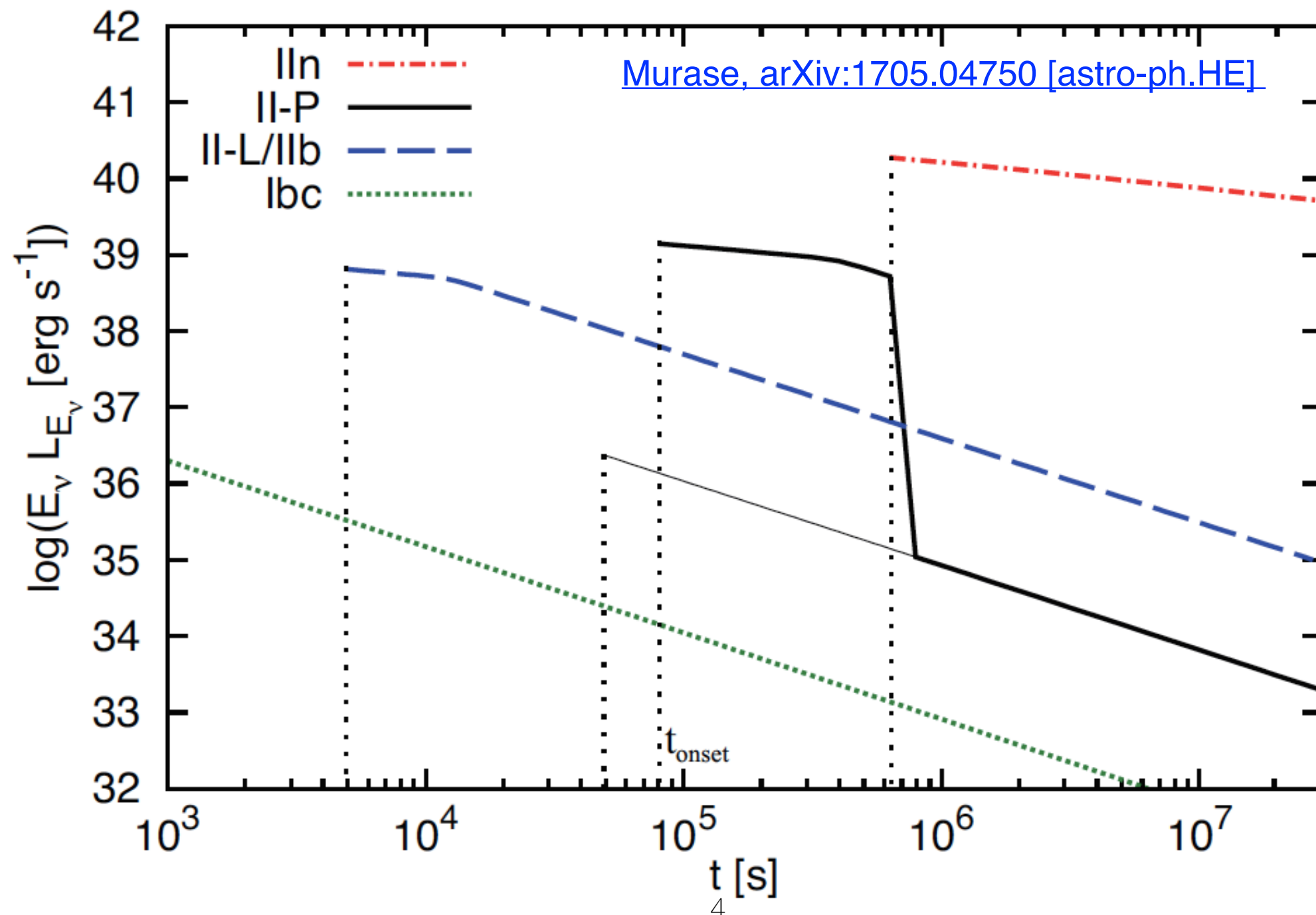
SNe candidates for HE neutrinos

SN candidates for the production of high energy neutrinos:

SNII-P (52.5% of SN type II) (Plateau in the light spectrum).

SNIIIn (6.75% of SN type II) (narrow emission lines), SCM very dense.

Rare, but can be detected up to ~ 10 Mpc.



CCSN neutrino flux / Expected neutrino rate

• SNe HE neutrino energy fluence

The neutrino energy The neutrino flux The effective pp optical depth set to 1 for SNIIP and SN2n (depend on the cross section and proton inelasticity) The energy fraction carried by cosmic rays = 0.1 The proton momentum index (2.0 - 2.2)

$$E_\nu^2 \phi_\nu \sim \frac{1}{4\pi d^2} \frac{1}{2} \min[1, f_{pp}] \frac{\epsilon_{cr} \epsilon_s}{R_{cr10}} \left(\frac{E_\nu}{0.4} \right)^{2-s}$$

The flux decreases with d over the surface area of a sphere

$R_{cr10} \equiv \epsilon_{cr} \epsilon_s / (E_p^2 dN_{cr}/dE_p)|_{10\text{GeV}}$

The proton energy The number of cosmic rays

The bolometric cosmic ray energy is divided by the differential cosmic ray energy at 10 GeV

$\epsilon_s \approx L_s t_{dyn}$

The kinetic luminosity The dynamical time

$L_s = 2\pi \rho_{cs} V^3 R_s^2$ $t_{dyn} \approx \frac{R_s}{V_s}$

The density profile of the CSM The velocity of the shell

$V_s = \frac{dR_s}{dt}$ The shock radius

• Expected number of neutrino observable in KM3NeT

$$N_\nu = \int_{t_{min}}^{t_{max}} \int_{E_{min}}^{E_{max}} \phi(E_\nu, t) A_{eff}(E_\nu) dt dE_\nu$$

$[t_{min}, t_{max}]$: the observation time

High energy CCSN KM3NeT analysis

1. Search for neutrino emission from individual supernovae

To find an excess of neutrino from given SN positions and times

-> Use a **time dependent** point source likelihood method.

[IceCube paper\(2023\)](#)

The number of
neutrino events

The number of
fitted signal events

Signal PDF

Background PDF

$$\mathcal{L} = \prod_{i=1}^N \left(\frac{n_s}{N} \mathcal{S}(\nu_i) + \left(1 - \frac{n_s}{N} \right) \mathcal{B}(\nu_i) \right)$$

Signal and background probability distribution functions (PDFs):

Spatial term $\times$ Energy term $\times$ Time term $\longrightarrow$ **Currently I start using the aanet unbinned framework**

$$\mathcal{S} = \mathcal{N}_S \times \mathcal{E}_S(\gamma) \times \mathcal{T}_S$$

The Signal time PDF == Box function / Function: neutrino light curve.
The Background time PDF: data driven background.

The expected neutrino energy spectrum:

- The signal energy PDF follows ($E^{-\gamma}$).
- The Background energy PDF: follows the distribution observed in the data.

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Implementing in
aanet

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Signal and background time PDFs

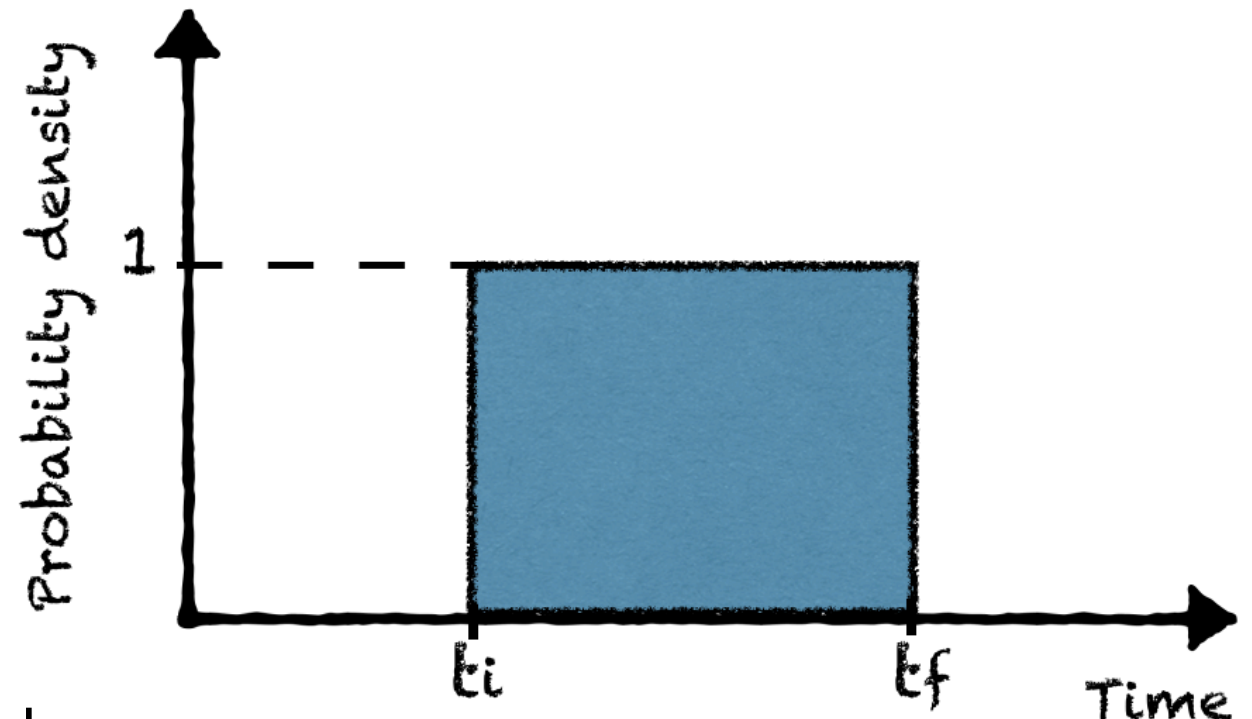
- Signal time PDF:

1) **Box function** implemented.

$$= 1 \text{ if } (t_i < t < t_f)$$

t_i : start time of emission

t_f : end time of emission



2) **Function = Light curve** to be implemented.

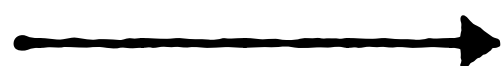
High energy supernova neutrino light curves:

$$\mathcal{T}(t) \propto (1 + t/t_{pp})^{-1}$$

t_{pp} the characteristic time scale constant

3) **Light curve files** generated with gSeaGen (from Murase fluxes) to be implemented.

- Background time PDF:



Follow data time distribution

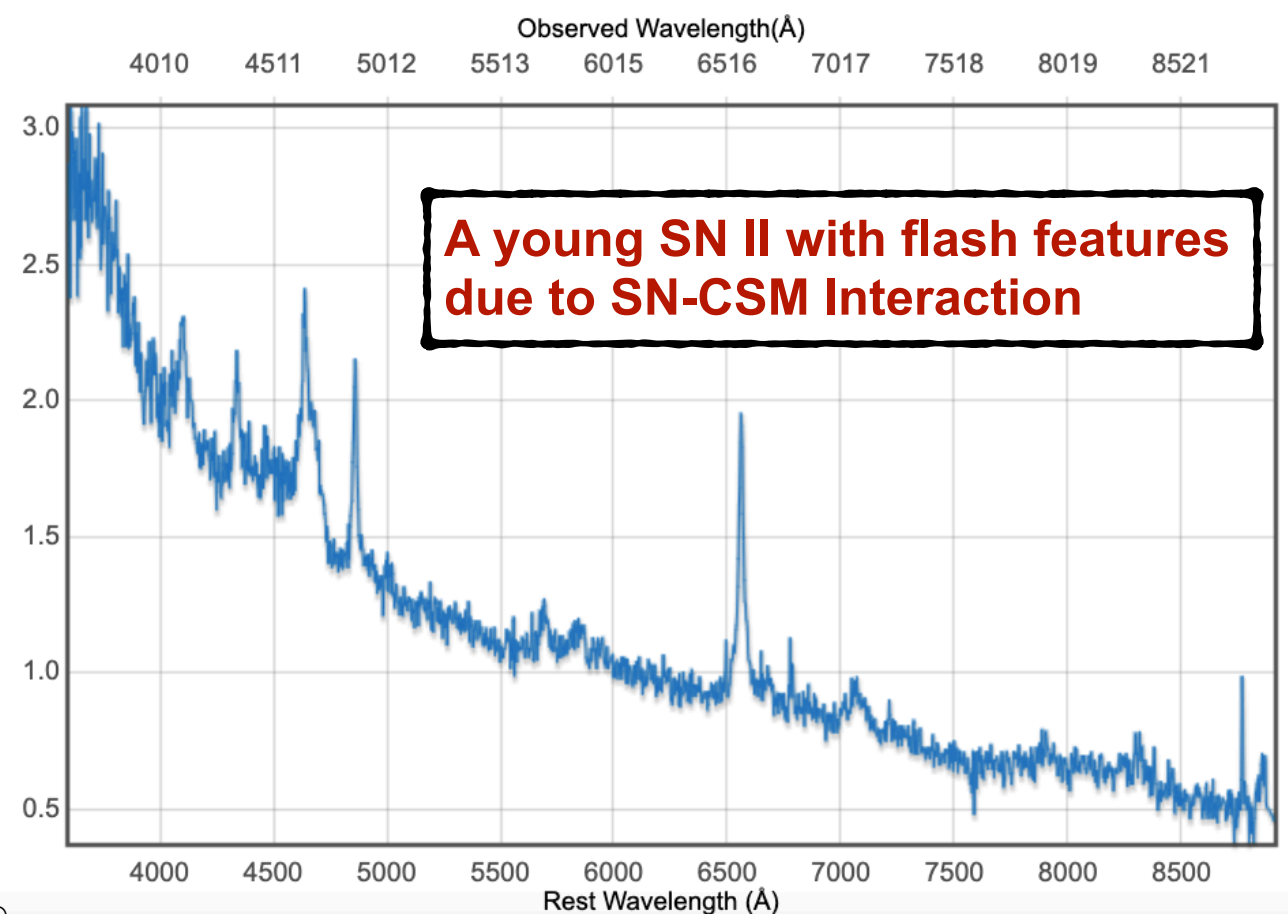
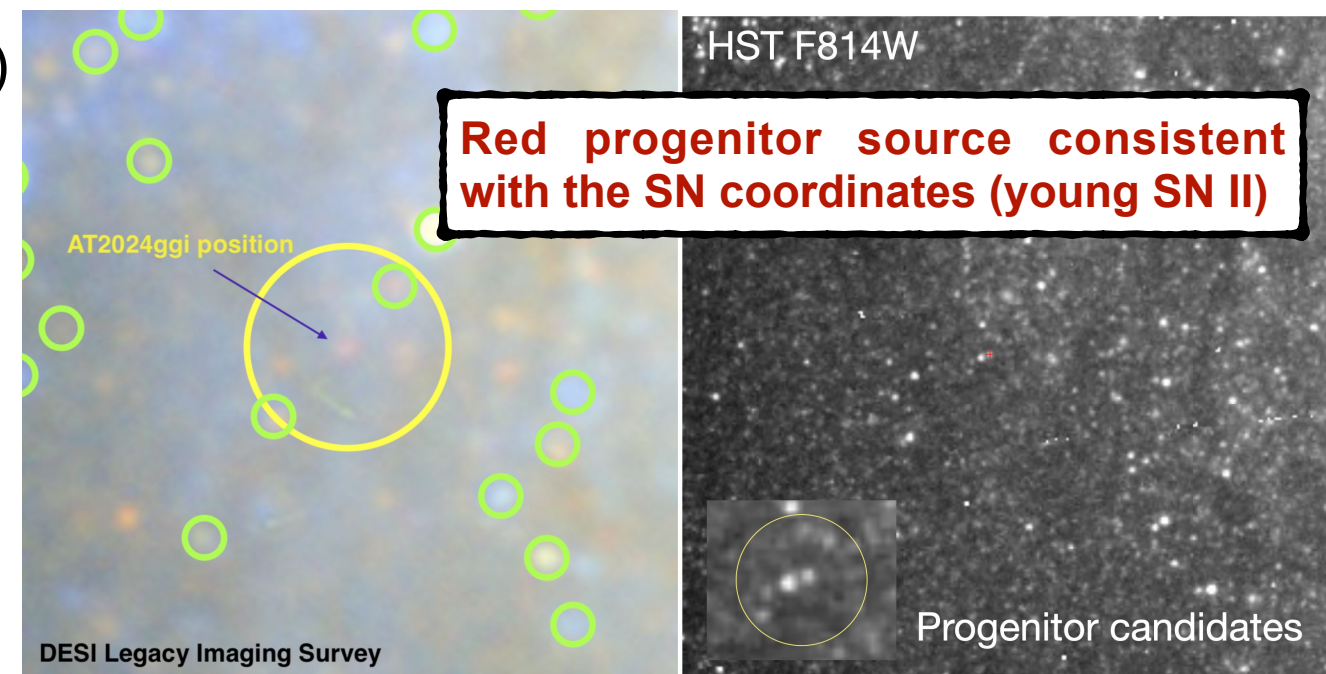
SN2024ggi candidate

Discovery date in MJD: 60411.14 (2024-04-11.14)
Location: NGC3621 (at ~ 7 Mpc)
RA : 11:18:22.087 / 169.592030106
DEC: -32:50:15.26 / -32.8375716158
Redshift: 0.002435

Atel #16587: NuSTAR Observations of SN2024ggi 2 days post discovery

*" we find clear evidence for hard-X-ray emission at the location of the supernova with statistical significance > 10 sigma in each of the two NuSTAR modules. A preliminary spectral analysis shows that the data can be modeled with an absorbed power-law spectrum with photon index $\Gamma \sim 1.4 \pm 0.2$ and intrinsic $N_H < 3 \times 10^{23} \text{ cm}^{-2}$ in the energy range 3-40 keV. **The inferred absorbed flux is $\sim 3 \times 10^{-12} \text{ erg/s/cm}^2$ in the 3-80 keV energy range, corresponding to a luminosity of $\sim 2 \times 10^{40} \text{ erg/s}$ at the distance of 6.6 Mpc. The hard photon index is suggestive of an intrinsic thermal bremsstrahlung spectrum. Assuming a thermal bremsstrahlung spectral model we infer $T > 40 \text{ keV}$. Hot bremsstrahlung emission is expected to originate from **the SN shock interaction with a dense medium.** "***

<https://www.astronomerstelegam.org/?read=16587> 9

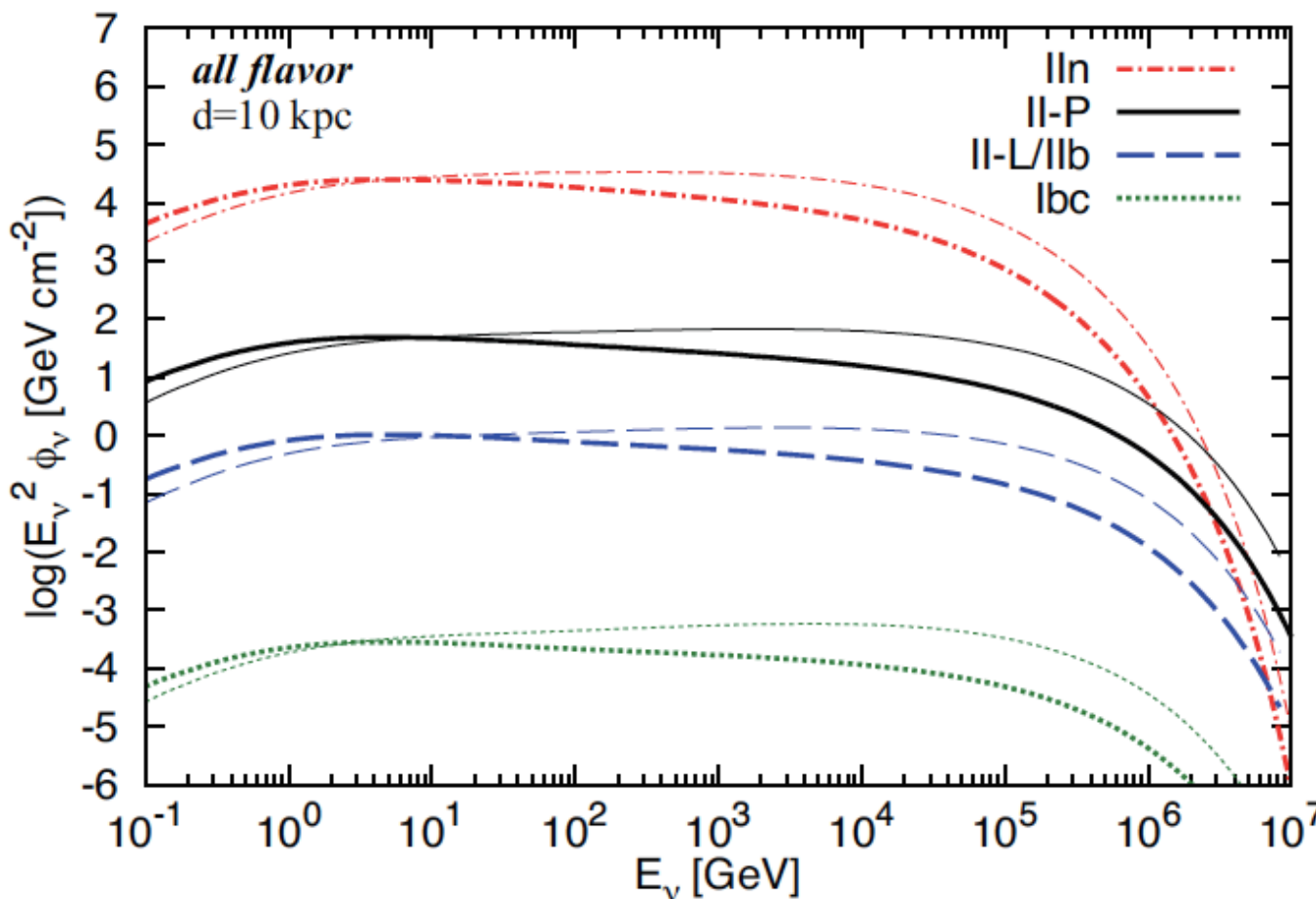


SN2024ggi candidate

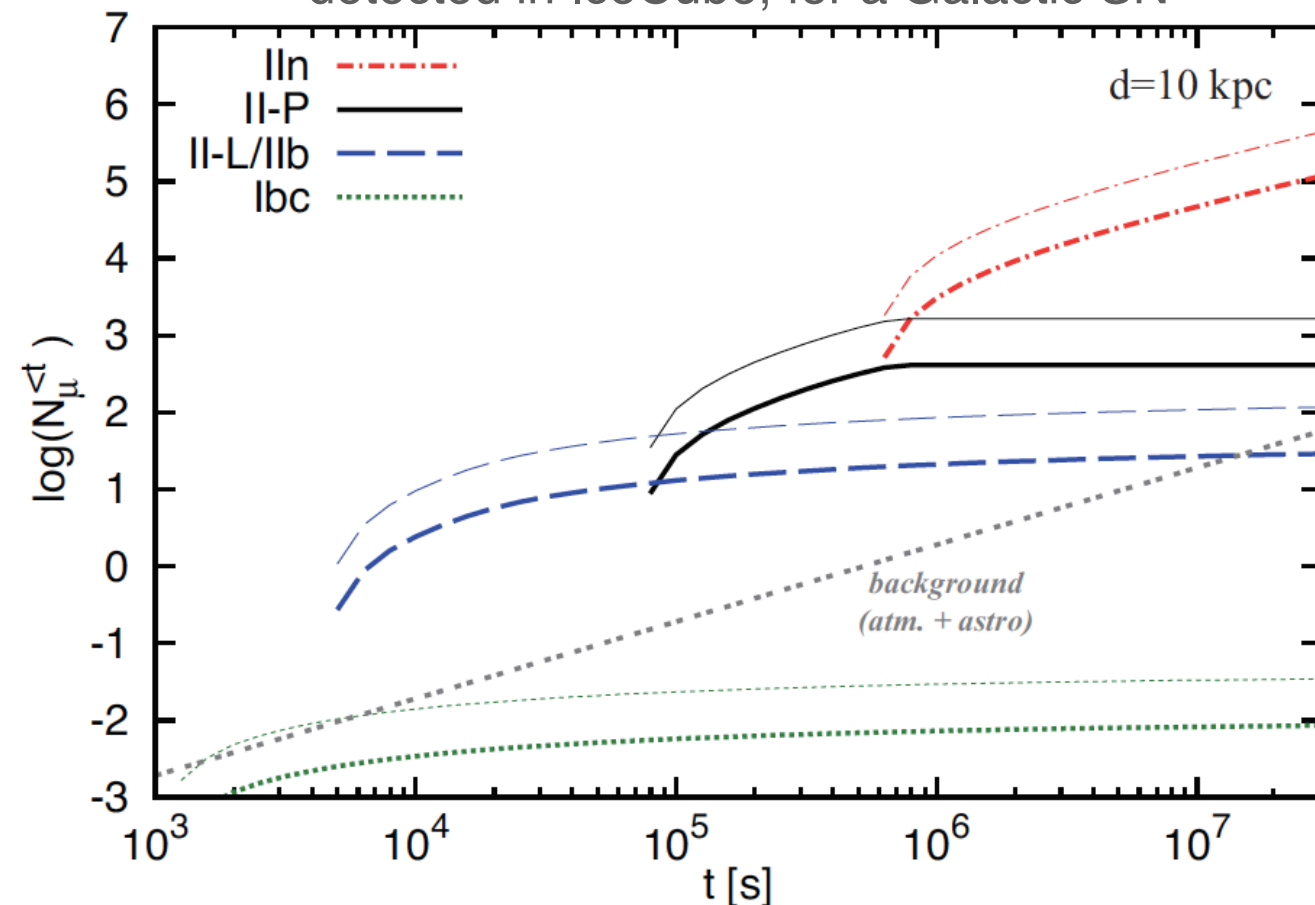
Possible detection of SN2024ggi neutrinos with KM3NeT?

- For a Galactic SN, Gton neutrino detectors like IceCube and KM3Net can detect > 10 HE neutrinos from a Type II SN.
- 10^3 and 10^6 events are possible for Type II-P and IIc SNe
- The targeted time window is . 0.1-1 day after the MeV neutrino detection for SNe II-P/II-L/IIb (SNe IIc) 0.1-1 yr for SNe IIc.
- The expected emission span from 1 GeV up to 10 PeV.

Energy fluences of from a Galactic SN



Expected numbers of through-going muon tracks detected in IceCube, for a Galactic SN



2) High energy CCSN KM3NeT analysis projects

In addition to probing the neutrino fluxes from single SNe:

→ combine the signal of a sample of SNe with a **stacking analysis**.

Motivation for a stacking analysis:

The dataset is highly background dominated, see [IceCube paper\(2023\)](#)

2. Search for a combined emission from a supernova sample through a time-dependent **stacking analysis**.

Implement a source stacking through a weighted sum of the signal PDFs S_i of individual S_j :

$$\mathcal{S} = \sum_j w_j \mathcal{S}_j$$

The expected signal strength of the sources

$$w_j \propto \underbrace{\frac{\Phi_0}{D_p^2}}_{\text{Source Properties}} \times \underbrace{\int_{t_{\text{start}}}^{t_{\text{end}}} \int_{E_{\text{min}}}^{E_{\text{max}}} L_j^\nu E^{-\gamma} A_{\text{eff}} dt dE}_{\text{Time Dependence}}$$

Φ_0 : the intrinsic neutrino power of the sources

D_p^2 : the proper distance of the SN

L_j^ν : the estimated neutrino light curve

$E^{-\gamma}$: the neutrino energy spectrum

$A_{\text{eff}}(t, \delta_j, E)$: the effective area

E : the energy

δ : the declination of the source